

# O&M Cane Training App

Project status report · 6 July 2026 · Prepared for management review

## 1 · Executive summary

The app — an offline-first Android tool that lets teachers at three partner schools for the blind run structured cane-training assessments with visually impaired children — is **feature-complete for single-device pilot use and demo-ready today**. The last working session closed out the entire engineering side of data-protection compliance (guardian consent capture, evidence, and erasure), fixed two real bugs found by a full code audit, and stood up the cloud backend. The next milestone is connecting the app to that backend, which turns the single-phone pilot into a multi-device, centrally-synced pilot. The remaining blockers are largely **inputs from other teams** (legal sign-off, translations, teacher roster), not engineering.

## 2 · What the product is

Teachers select an activity from a structured curriculum, pick a child, run the assessment, and record scores — with optional video evidence of the child performing the activity. Reference material (demonstration video, standard operating procedure, facilitator notes, multilingual audio narration) is one tap away on every screen. Everything works fully offline; data lives on the device and exports to CSV.

Four design decisions define the architecture:

|                        |                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Offline-first          | Pilot schools cannot rely on connectivity. Every assessment works with zero network; the cloud is an addition, never a dependency.                                          |
| Pseudonymous by design | Each child receives a research ID at enrolment. All records, video files and research exports carry only this ID — names and dates of birth never leave the identity layer. |
| Consent-gated video    | Video capture is locked until a verifiable guardian consent record exists; withdrawal and erasure are first-class flows, not afterthoughts.                                 |
| No build step          | The app is deliberately plain HTML/CSS/JS (wrapped as a native Android app), so the content team can edit activity text and rubrics directly without any developer tooling. |

## 3 · Where the project stands

| Area                                       | Status  | Notes                                                                                                                                                                   |
|--------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core app (activities, assessment, records) | Done    | Full flow shipped and verified on device: login, activity catalogue, child picker, record screen, past results, CSV export.                                             |
| Video evidence capture                     | Done    | Clips stored on-device keyed by research ID; memory-safe for long clips; failures reported honestly to the teacher.                                                     |
| Consent & compliance engineering           | Done    | Verifiable consent envelope, photographed proof of the signed form, withdrawal and full erasure. Closed last session.                                                   |
| Compliance documentation                   | Drafted | Privacy policy, guardian consent form (with translation draft), app-store data-safety answers, and a requirement-by-requirement compliance map — awaiting legal review. |
| Cloud backend                              | Live    | Managed Postgres project created in an India data region; full schema deployed (schools, teachers, children, records) with per-school access isolation.                 |

| Area                   | Status  | Notes                                                                                                         |
|------------------------|---------|---------------------------------------------------------------------------------------------------------------|
| App ↔ cloud connection | Next up | Login against real accounts, server-assigned child IDs, video upload. Design finalised; code not yet written. |
| App-store release      | Pending | Data-safety answers drafted; needs the hosted privacy policy and the cloud connection first.                  |

## 4 · Problems solved in the last working session

The session had one theme: **make the app defensible before it touches children's data at scale** — finish consent, prove the code is sound, and lay the cloud foundation.

### ■ Guardian consent, finished to a legal standard

What existed was a bare checkbox. It is now a verifiable consent record: the guardian's name, relationship to the child, and method of consent are required before video capture unlocks. Withdrawal is audit-honest — it stamps a withdrawal date but preserves the original grant (history is never rewritten), and prompts the teacher to delete stored clips if the guardian requests erasure.

### ■ Proof that consent existed

Teachers can now photograph the signed paper consent form and record its serial number inside the app. The evidence is stored against the child's research ID, viewable by an auditor, deliberately survives consent withdrawal (it proves past processing was lawful), and is destroyed when the child's data is deleted.

### ■ Complete erasure, verified

Deleting a record, a child, or all data now removes video *files* from disk, not just the pointers to them. No orphaned children's data can remain on a device.

### ■ Independent full-code audit — two real bugs caught

The whole codebase was audited with the new code treated as suspect. Two genuine consent-flow bugs were found and fixed — including one where consent evidence could attach to the *wrong child*, which is worse than having no evidence at all. The audit's overall verdict: well-engineered for a pilot, unusually strong compliance posture and documentation.

### ■ Long-video memory fix

Storing a multi-minute clip previously risked crashing the app by loading it into memory at once. Clips are now copied in small slices — memory use is constant regardless of clip length — and a failed copy cleans up after itself instead of leaving a corrupt file.

### ■ Cloud backend stood up

The managed database project is live in an India data region (a deliberate data-residency choice), with the full schema deployed: per-school data isolation enforced at the database layer, and a server-side enrolment routine that mints each child's research ID centrally.

### ■ A subtle cross-cutting bug caught before it shipped

The database was seeded with school identifiers in a different format than the app uses. Because security rules match on that identifier, the mismatch would have silently rejected every read and write once connected — a class of bug that is miserable to diagnose later. It was caught in review; the fix is decided and applied to the schema.

## ■ Research design locked

The four open research questions were answered and locked: child identity is server-assigned at enrolment (one moment of connectivity per child, once); video evidence is confirmed as required; the same child may be assessed from multiple devices; and analysis will be both longitudinal and cross-child. This unblocked the single biggest rework risk on the plan.

## 5 · What happens next

Engineering work, in order of rework risk:

| # | Work item                                       | What it delivers                                                                                                                                                                                                     |
|---|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Backend configuration &amp; test account</b> | Verify the deployed schema, apply the identifier fix to the live database, create the private video-storage bucket, and provision one test teacher account. Roughly ten minutes of dashboard work; already scripted. |
| 2 | <b>Connect the app to the cloud</b>             | Real login, server-assigned child IDs at enrolment, offline behaviour preserved. Built behind a feature flag that is OFF by default — the current offline pilot is untouched until deliberately switched on.         |
| 3 | Video upload to cloud storage                   | Consent re-checked before every upload; erasure honoured server-side (delete-everywhere).                                                                                                                            |
| 4 | Real-device verification                        | Full pass on physical hardware, including the video-picker path that emulators handle differently.                                                                                                                   |
| 5 | App-store internal release                      | Invite-only distribution track as the pilot's real access control; target audience declared 18+ (teachers are the users; children are data subjects, not users).                                                     |

Waiting on inputs from outside engineering:

|                           |                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Legal team</b>         | Entity of record and grievance officer for the privacy policy; sign-off on the consent due-diligence process; answer on the educational-institution exemption. All questions are pre-packaged in the compliance map document. |
| <b>Content team</b>       | Verification and typesetting of the translated consent form; translated activity scripts (the multilingual audio pipeline is ready and waiting on this text).                                                                 |
| <b>Pilot coordination</b> | Real teacher roster for the three schools, so live accounts can replace test ones.                                                                                                                                            |

## 6 · Risks being watched

|                                                |                                                                                                                                                                                                                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Single-device data</b>                      | Until cloud sync ships, each phone is the only copy of its data. Mitigation in place: regular CSV exports and the audit confirmed export paths are sound. Resolved permanently by item 2 above.                                             |
| <b>Human dependencies on the critical path</b> | Legal and translation inputs gate the consent paperwork and the app-store listing. Engineering is sequenced so none of it blocks code progress.                                                                                             |
| <b>Server-side child names</b>                 | To let a second device show the child picker, names are stored server-side behind per-school access isolation, while research extracts remain pseudonymous. This is flagged for legal review and reversible until the first real enrolment. |

Bottom line: the app is demo-ready on a phone today. The cloud-connected, multi-device pilot is the next milestone, and it is gated mainly on inputs from other teams rather than on engineering.